



COURSE HANDOUT (Part II)

Date: 2/08/2026

In addition to Part I (General Handout for all courses appended to the Time Table), this portion gives further specific details regarding the course.

Course No. : CHE F411
Course Title : Environmental Pollution Control
Instructor-in-Charge : **Banasri Roy**

1. Course Description

Air & water pollutants; sampling and analysis; control methods for air & water pollutants; modeling of different control techniques; advanced wastewater treatment processes; solid waste management, noise pollution; case studies; associated laboratory.

2. Scope and Objective

1. Students will learn significance of environmental engineering course, effect of greenhouse effect on climate change, river pollution etc.
2. They will know theoretically how to do air pollutants sampling & Analysis of Air Pollutants
3. They understand basic modelling techniques of the dispersion of the pollutant in air.
4. They will learn basic designing aspects of different methods and equipment for air pollution control.
5. Students will gather idea on sources & classification of water pollutants: Origin of wastewater, their effects, Dissolved oxygen estimation, etc.
6. They learn how to collect wastewater samples and analyse: Understanding and analysis of concepts of DO, BOD, COD, TOC, Water Quality Standards, etc.
7. They comprehend the basics of wastewater treatment methods, biological treatments, Tertiary treatments, Zone settling process & technology
8. Students will learn Sources and Classification of solid wastes, Methods of Collection, Disposal methods, different Hazardous Wastes, treatment and managements, etc.
9. In Module 9 they will explore sources of noise, effects, measurement of noise, control techniques, etc.

3. Course Learning Outcome (CLO)

CLO 1: At the end of this course student should be able to Analyze the challenges of and suggest basic methods to control for the following items

- a) Air borne pollutants.
- b) Water borne pollutants.
- c) Solid waste pollutants.
- d) Noise pollution.
- e) Hazardous pollutants.

CLO 2: At the end of this course student should have the preliminary understanding/skill set required to work on designing and/or capacity enhancement of

- a) Air pollution control equipment.



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- b) Water treatment processes.
- c) Solid waste disposal and control processes.

CLO 3: At the end of this course student should have the preliminary understanding/skill set required to apply Laws and regulation related to pollution control in general.

4. Text Book

- T1.** Davis, M. L. and D. A. Cornwell, "Introduction to Environmental Engineering", McGraw Hill, New York, 5th ed., 2013.
- T2.** Rao, C.S., "Environmental Pollution Control Engineering," New Age International 2nd Ed., 2006.

5. Reference Books

1. R1. Sincero A. P., Sincero G. A., "Environmental Engineering : A Design Approach," PHI, 1996.
2. R2. Peavy, H.S., Rowe, D.R. and Technobanolous, G., "Environmental Engineering" McGraw Hill, 1985.
3. R3. Metcalf & Eddy, "Wastewater Engineering : Treatment and Reuse", 4th Edition,
4. Research papers from different journals.

6. Course Plan

In addition to the textbooks and reference books, different articles/reports will be provided in class in a regular manner on different topics.

Module No & Topic	Lecture Session	Reference	Learning Outcome (Students will be capable to-)
1. Introduction , Air pollution: sources & effects	L1: Introduction, global warming, greenhouse effect, climate change, river pollution etc. Significance of Environmental Engineering course, Handout discussion	Ch. 9, T1 Ch. 1, T2	Analyze/understand different types of air pollutants, their sources & effects. Develop awareness about the air acts
	L2: Definition and Scales of Concentration, Classification and Properties of Air Pollutants Emission Sources	Ch. 2, T2	
	L3: Effects of air pollution, The air act	Ch. 2, T2	
2. Air pollution sampling & measurement	L4: Types of pollutant sampling and measurements, Ambient Air Sampling	Ch. 3, T2	Analyze the types of air pollutants sampling & measurement.
	L5: Stack Sampling, Analysis of Air Pollutants		
3. Meteorologic al aspects of air pollutant dispersion	L6: Temperature Lapse rate and Stability	Ch. 9, T1 Ch. 4, T2	Analyze/understand meteorological conditions prevailing in the atmosphere, which
	L7: Wind Velocity and Turbulence, Plume Behavior		



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	L8: Dispersion of Air Pollutants L9: Estimation of Plume Rise L10: Application of Gaussian Plume model for estimating pollutant concentration at any place		affect the dispersion of pollutants emitted into the atmosphere.
4. Air pollution control methods and equipment	L11: Types of control, efficiency calculation, various types of particulate emission control methods L12: Gravity settling chamber, cyclone separator L13: Bag filter, electrostatic precipitator L14: Gaseous pollutant removal: Absorption L15: Gaseous pollutant removal: Adsorption, combustion L16: Gaseous pollutant removal (centrifugal and venturi scrubber) L17: Control of specific gaseous pollutants (SO _x , NO _x , CO)	Ch. 9, T1 Ch. 5 & 6, T2	Analyze/understand control methods for removal of particulates from air.
5. Sources & classification of water pollutants	L18: Water Resources, Origin of wastewater, Types of Water Pollutants and their effects L19: Dissolved oxygen estimation, DO sag curve L20: Ultimate BOD determination using Thomas method L21: Types of water pollutants, The water act	Ch. 7, T1 Ch. 7, T2	Analyze/understand the different types of water pollutants & their effects.
6. Wastewater sampling and analysis	L22: Sampling, Method of Analysis (Understanding of concepts of DO, BOD, COD, TOC) L23: Analysis of DO, BOD, COD, TOC L24: Determination of Inorganic Substances, Physical Characteristics, Bacteriological Measurements, Water Quality Standards	Ch. 8, T2	Analyze/understand the different methods of analysis of water pollutants.



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7.Wastewater treatment	L25: Basic Processes of Water Treatment, Pretreatment, Screening, Sedimentation,	Ch. 9, T1 Ch. 9, T2 And Guest Lecture by Industrial Person	Analyze/understand the primary, secondary and advanced treatment methods for water and wastewater.
	L26: Discrete settling phenomenon, design of sedimentation tank		
	L27: Biological treatment, study and design of activated sludge process, ASP for BOD removal		
	L28: N & P removal from wastewater -MLE & Bardenpho processes, microstraining, adsorption		
	L29: Tertiary treatments for reuse : - Rapid Gravity Filters; Ultra Filtration; -Reverse Osmosis		
	L30: Zone settling process & technology : - Primary clarifier lamella clarifier technology, design of secondary clarifier, suction type clarifiers technology		
8.Solid and Hazardous waste management	L31: Solid waste: Sources and Classification, Public Health Aspects, Methods of Collection, Disposal Methods, Potential Methods of disposal	Ch.11, T1 Ch.10, T2	Analyze/understand the sources of solid wastes, their types, effects, methods of collection & methods of disposal. To study hazardous waste management including radioactive waste management.
	L32: Hazardous Wastes : Definition and Sources, Hazardous Waste Classification	Ch.12, T1	
	L33: Potentially Non Compatible Wastes, Hazardous Waste Management Strategy, Treatment Methods, Solidification and Stabilization		
	L34: solid waste management regulation, hazardous waste management regulation		
	L35: Basics of radioactivity & radiation, effects of radiation and protection methods	Ch. 11, T1	
	L36: Types of radioactive wastes, disposal methods	Ch. 11, T1	
9. Noise pollution	L36-38: characteristics of sound wave, sources of noise, effects	Ch.10, T1	Analyze/understand the sources of noise



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	L39-40: Measurement of noise, control techniques, noise regulation acts		pollution, its effects, measurement of noise & control techniques for noise pollution.
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7. Evaluation Scheme

EC No.	Evaluation Component (EC)	Duration	Weightage %	Date/Time	Remarks
1	Announced Quiz (6 out of 8)		15	Continuous mode	OB/CB
2	2 Assignments (To be discussed/announced in class)		20	-	OB
3	1 Seminar (To be discussed/announced in class)		10		OB
4	2 Research paper study/case study etc. (To be discussed/announced in class)		15		OB/CB
5	Comprehensive Exam	3 hrs	40		Close Book

Student will be graded based on their performance on

1. Solving numerical, short conceptual, MCQ problems at the level of the prescribed text books and content covered. Detailed will be discussed in class.
2. Assignments/Seminar/research paper study/case study etc.

8. IMPORTANT NOTES

- **Chamber consultation hour** will be announced in the class.
- The **notices/announcement**, if any, concerning the course will be discussed/displayed on class, and/or Google Class, and/or the NALANDA only.
- **Make-up** will be granted for **genuine cases only**. Certificate from authenticated doctor, say from the Medical Center, must accompany make-up application (*only prescription or vouchers for medicines will not be sufficient*). Prior permission of IC is compulsory.
- **No make-up will be granted for the EC Nos. 1 to 4.**

Instructor-in-charge | **CHE F411**